Does aging microplastics alter how particles affect the physiology of a freshwater copepod

The Zooplankton Ecology & Adaptation Lab At UMass Lowell

Jackson Vanegas, Matthew Sasaki

UML Biology

Introduction

- Microplastics are increasing in abundance in aquatic systems
- Changes in food quality and quantity
- Copepods are important, highly selective aquatic grazers
- Biofilms can form on surface of microplastics
  - Nutritional benefit?
  - Increased exposure via reduced selectivity?

Central Question

Does microplastic aging alter how these particles affect the physiology of a common freshwater copepod?

Hypothesis

1. Exposure to microplastics will decrease the thermal limits in the freshwater copepod *Skistodiaptomus pallidus.*
2. The upper thermal limit for copepods exposed to aged microplastics will be higher than the upper thermal limit for those exposed to un-aged microplastics.
3. Because of the nutritional content of biofilms, starved copepods with aged microplastics will exhibit a smaller decrease in thermal limit relative to the control copepods, while starved copepods with un-aged microplastics will exhibit a sharp decrease in thermal limits.

Preliminary Results

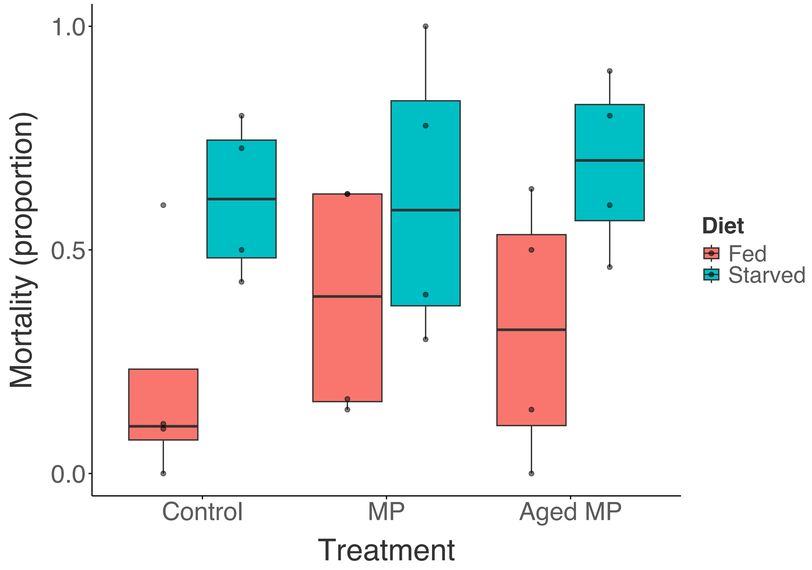


Figure 1: Descriptive figure caption here

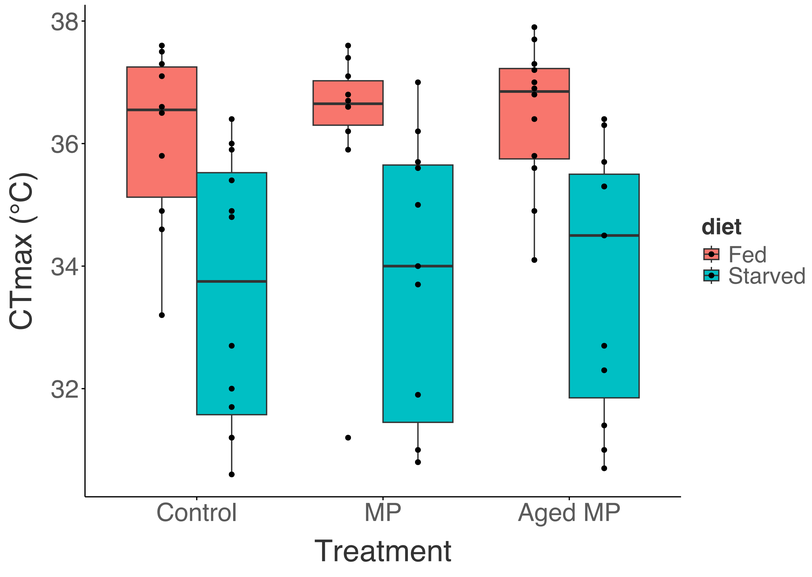
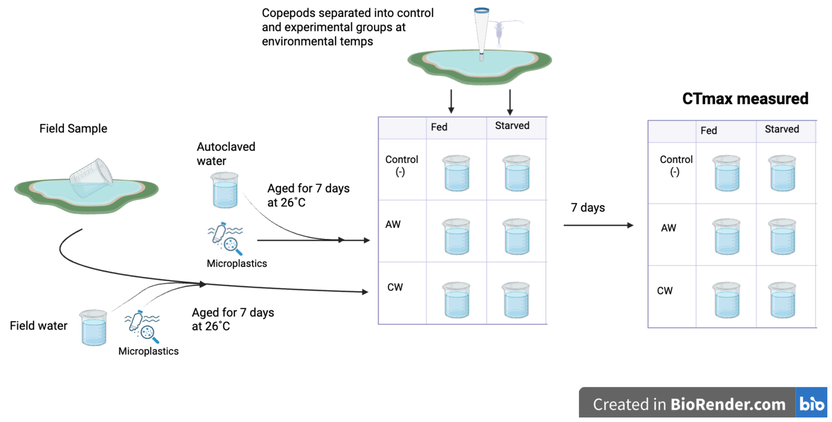


Figure 2: Descriptive figure caption here

Methods



Conclusion

- Starvation has major effects on both survival and thermal limits
- Realistic microplastic concentrations have minimal effects on thermal limits, but may influence survival
- Survival seems to be highly variable from week to week

Next Steps

- Additional experimental replicates
- Dose-dependence of effects

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